STRIDE: Static-profile Routing with Invariant Dynamics Encoding

Cross-room few-label WiFi-CSI activity recognition — CS60055 Hackathon 2

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Abstract

We build a fully Python pipeline from raw Nexmon PCAP (Room A) and amplitude CSV (Room B) to room identification, self-supervised representation learning and few-label cross-room activity recognition. Inspecting the data rather than assuming its format exposed three facts that shape everything downstream: (i) half of the Room-A packets are a fixed, activity-independent pattern; (ii) both rooms mix 20 MHz frames into the 80 MHz CSI stream, at a rate that in Room B depends on the recording session (and hence on the activity); (iii) Room A never yields 150 valid observations per capture, while Room B is sampled roughly four times faster. We remove invalid observations with physical criteria that are identical for both rooms, build windows with one symmetric rule, and decompose log amplitude into a static multipath profile (used for routing) and a low-rank-denoised dynamic part (used for activity). On the fixed validation sets, the router identifies the room with accuracy 1.000 (0.949 from a single observation), self-supervised pretraining raises the linear-probe macro-F1 well above the handcrafted and random-weight baselines, and cross-room few-label adaptation reaches a macro-F1 of 0.341 (B→A) and 0.418 (A→B) at K=20. We report what does *not* transfer (SIT vs STAND) and why.

1 Problem understanding

A deployment in a new room can afford only a few labelled captures. The benchmark has two rooms recorded with the same BCM4366c0 hardware at 80 MHz but in different sessions. Task 1 identifies the room from a $(150, 256)$ amplitude window; Task 2 learns a representation without labels; Task 3 recognises the target room's own activities from $K \in \{1, 5, 10, 20\}$ labelled captures each, and the shared SIT/STAND/WALK activities from the *other* room's labels only. The capture is the atomic unit; validation is never used for fitting; no metadata (format, file name, length) may act as a feature. We treat these rules as engineering constraints and enforce them in code (split assertions, a development-only deployment artefact, random window ids, format discarded before routing).

2 Data engineering

Decoding. We parse classic libpcap records, locate the UDP payload and the Nexmon header (magic `0x1111`, sequence, core/stream, chanspec `0xe02a`=80 MHz), and unpack each 32-bit BCM4366c0 word as sign/12-bit mantissa for I and Q plus a shared 6-bit exponent, following the public Nexmon CSI format description [2] (the authors' MATLAB code was never opened). Correctness is tested against physics, not against another implementation: synthetic words round-trip, real spectra are far smoother across adjacent subcarriers than a shuffled order, the 80 MHz guard subcarriers (FFT bins 123–133) are deep nulls, and walking varies more over time than standing. Both rooms turn out to be in the same FFT-bin order.

Three kinds of observation. Figure 2 shows what the inspection revealed. Room-A reports strictly alternate between a normal sequence number and `0xFFFF`; the latter carry the same profile in every capture (mean cross-capture correlation 0.99 versus 0.22 for the normal reports), are activity-independent, about 34× weaker and lack guard nulls, so we reject them as zero-information

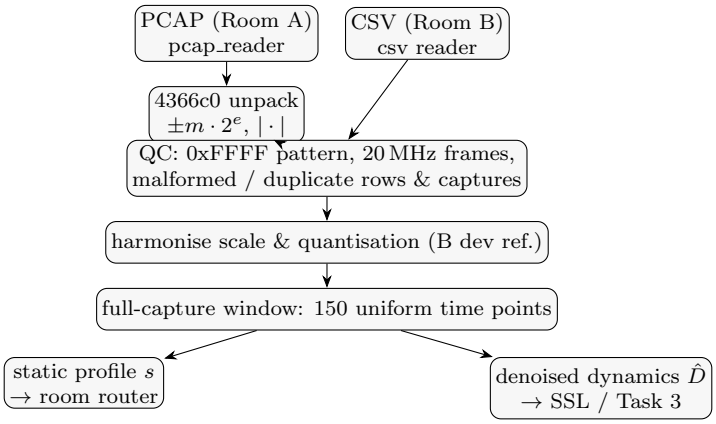


Figure 1: Pipeline (all stages are our own Python code; the source format only selects the reader).

(118,013 of 234,239 decoded reports). Both rooms also contain *narrow-band* frames whose energy occupies one 20 MHz quarter of the band; we reject an observation when fewer than three quarters carry energy (47,359 in A, 39,196 in B). This matters beyond cleanliness: in Room B such frames make up 12.7 % of SIT and 14.0 % of STAND rows, 6.8 % of LYING, but only 1.4 % of WALK and 0.1 % of EMPTY rows, a session artefact any model would happily learn as an "activity" cue. A byte-identical duplicate capture in the TA package (`stand65(1).pcap`) is removed by content hash.

Table 1: Quality control over all captures (both splits; the same rules are applied to every capture).

	Room A (PCAP)	Room B (CSV)
decoded observations	234,239	600,500
rejected: seq. 0xFFFF fixed pattern	118,013	0
rejected: narrow-band (20 MHz) frame	47,359	39,196
rejected: duplicate / non-finite / zero	0	0
valid full-band observations	68,867	561,304

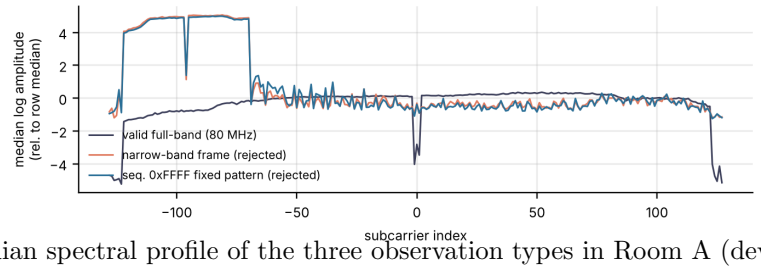


Figure 2: Median spectral profile of the three observation types in Room A (development data).

Harmonisation and windows. Decoded Room-A values have an arbitrary power-of-two scale and are non-integer, whereas Room B is integer; we rescale each Room-A observation to Room B's median level (estimated on Room-B development rows) and round, so neither scale nor quantisation can identify the room. After QC a Room-A capture holds a median of 59 valid observations (maximum 128), never 150. Room B is sampled faster: the lag-1 autocorrelation of the dominant motion components of WALK is 0.32 in Room A (native valid observations) and 0.40 in Room B, falling to 0.31 when Room B is decimated by four. Our deterministic rule, identical for both rooms, resamples the whole capture's valid sequence onto 150 uniform time points (timestamps for PCAP, row index for CSV) and rounds to the integer grid; one window is one complete recorded activity instance. A recorded ablation (`scripts/12`) against "150 consecutive valid rows when available" gives Task-3 macro-F1 0.300/0.366 (full) versus 0.303/0.316 (consecutive) for targets A/B, with shared-class F1 0.273 versus 0.204 on target B. No padding rows exist, so capture length cannot

leak through repetition; every window is traceable to its raw file, SHA-1 and configuration in a provenance manifest.

Splits. Room A uses the supplied validation folder. For Room B, candidates are the files matching the documented patterns, `sorted()` and sampled with `random.Random(42)`, 25 per activity (the four non-pattern files, which we verified are not duplicates, go to development). A stratified 20 % of development captures is held out internally for every tuning decision.

Table 2: Design decisions, the evidence behind them (development data unless stated) and the alternative we rejected.

Decision	Evidence	Rejected alternative
Reject seq. 0xFFFF reports	same profile in every capture (corr. 0.99), activity-independent, no guard nulls	keep all packets (as a naive reader would)
Reject narrow-band frames	energy in one 20 MHz quarter; class-dependent share in Room B (session artefact)	keep (gives a spurious SIT/STAND cue)
Harmonise Room-A scale & rounding	B is integer with a fixed level; A is arbitrary-scale float	leave as is (format becomes a room cue)
Full-capture windows, both rooms	A never has 150 valid observations; ablation: 0.366 vs 0.316 on target B	150 consecutive rows; padding (leaks length)
Low-rank denoising of D	naive motion energy inverted in Room B; +0.12/+0.24 shared F1 (Task 3)	raw D
Capture-consistency SSL, no reversal	capture = one activity; SIT-DOWN reversed = STAND-UP	time-reversal augmentation
Room-wise standardisation + LR head	best mean validation macro-F1 (Table 6)	prototypes; SAPT; balanced OT

3 Method

STRIDE decomposition. For a window A, $L = \log(A+1)$; subtracting each observation's median over data subcarriers removes the gain; the per-subcarrier median over time is the static profile s (environment, posture, location); the remainder D is the body-motion part. Motion changes many subcarriers coherently (low rank) whereas noise does not, so we keep only singular components of D above the Gavish–Donoho optimal hard threshold for unknown noise [3] (parameter-free, per window). This step proved decisive: in Room B the naive motion energy (median temporal std of D) is 0.79 for SIT and 0.77 for STAND (noisy sessions) but only 0.65 for WALK; after denoising WALK is highest (0.52 vs 0.43/0.46), restoring the physical ordering (Fig. 3).

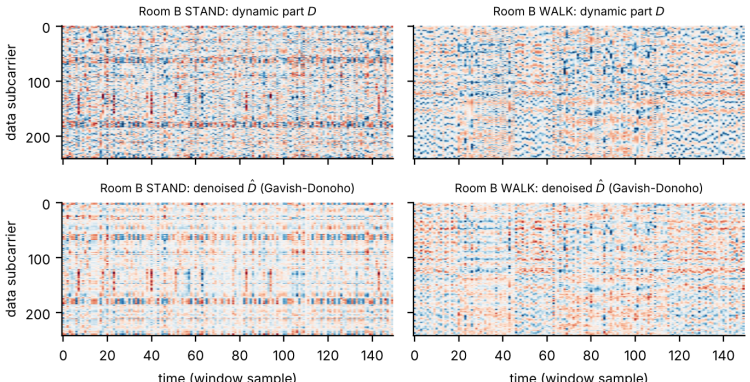


Figure 3: Dynamic part before and after low-rank denoising for two Room-B development windows. The noisy STAND session is dominated by incoherent fluctuations that disappear; coherent walking structure stays.

Task 1: router. An L_2-regularised logistic regression on s (C chosen on the internal hold-out). It works for any window length because s is a median over whatever observations exist.

Task 2: self-supervised encoder. A 121 k-parameter 1-D CNN: a 1×1 "spectral mixing" convolution over the 242 data subcarriers, four dilated depthwise-separable temporal blocks and mean‖std pooling to a 128-d embedding. It is trained with an NT-Xent loss [4] whose positives are two random time spans of the *same capture* — the only supervision is capture identity, no activity or room label. Augmentations: random time span (scale/shift), subcarrier-band and time masking, jitter; time reversal is deliberately excluded because SIT-DOWN reversed is STAND-UP. Two inputs are compared: the denoised dynamics $\hat{D}$ (S) and the gain-normalised log amplitude including the static profile (S_{raw}). Frozen embeddings are scored with a logistic-regression probe fitted on development windows, against handcrafted features (motion-energy bands, temporal spectrum and autocorrelation of the dominant motion component, static-profile PCs; on D (H) or $\hat{D}$ (H_d)) and the same CNN with random weights (R).

Task 3: routed, aligned few-label transfer. (1) The router selects the target room's label space and normalisation statistics. (2) Features are standardised per room using unlabelled development windows of that room (validation never). (3) A class-balanced logistic regression is trained on aligned source windows of the shared classes plus the K support captures of each target-exclusive class. Variants (ablations): cosine prototypes [5]; *SAPT* — statistics estimated only from shared-class windows (exact in the source, iteratively predicted in the target) to counter the label shift between rooms; balanced optimal-transport refinement on the unlabelled target pool [6, 7]; self-training; and a strict tier that uses no unlabelled target data. Hyper-parameters were fixed a priori (no target labels).

4 Experimental setup

All headline numbers use the fixed validation windows (150 Room-A, 125 Room-B captures, one window each). SSL: 60 epochs, batch 128, AdamW, 3 seeds, CPU only. Task 3: 10 nested support episodes (K=1 ⊂ 5 ⊂ 10 ⊂ 20 distinct development captures per exclusive class) for each encoder seed; we report mean±std over runs. Macro-F1 is the headline metric (balanced accuracy also reported); target-exclusive and shared macro-F1 are computed over the respective classes. The headline Task-3 configuration (room-align+LR on H_d features) was chosen by mean validation macro-F1 over both directions and all K; because validation guided this choice, all variants are reported.

5 Results

Task 1. Room accuracy is 1.000 at L=150, 0.985 at L=5 and still 0.949 from a single observation (Fig. 4). The audit is equally informative: motion energy alone gives 1.000 and a four-number temporal-texture probe gives 0.996, i.e. the rooms differ not only in multipath but in session properties (sampling rate, noise floor), while the quantisation probe is at 0.273 after harmonisation. We therefore read the router as a *recorded-environment* fingerprint (Table 3).

Table 3: Task 1 audit: room accuracy (validation, L=150) of the same classifier on different evidence.

Features	Accuracy
static profile s (256)	1.000
static profile, data bins (242)	1.000
motion-energy profile (242)	1.000
temporal-texture probe (4)	0.996
quantisation/level probe (3)	0.273

Task 2. Table 4: self-supervised pretraining clearly beats both required baselines. S_{raw}, which keeps the static profile, reaches the highest within-room and joint accuracy; removing the static

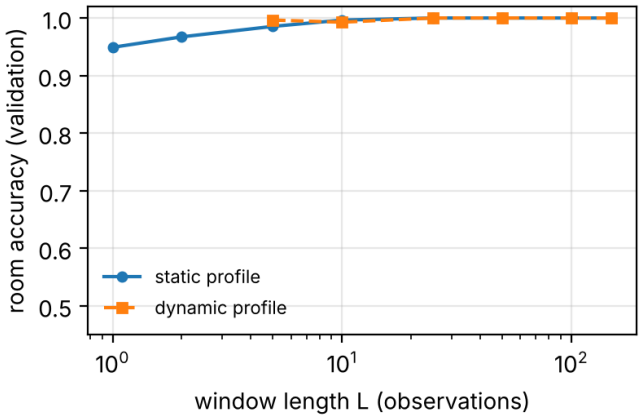


Figure 4: Task 1: room accuracy versus window length (same validation captures).

profile (S) costs Room-A accuracy, confirming that posture and location live largely in s. This is the invariance–discriminability trade-off we anticipated. Zero-shot transfer of a source-room SIT/STAND/WALK probe to the other room stays low for every representation, which motivates Task 3's alignment. UMAP plots (Fig. 8) show embeddings separating by room first — consistent with the Task-1 audit — and by activity within each room. In the STRIDE-SSL embedding the Room-A WALK windows lie next to the Room-B WALK windows, forming the bridge between the two room clusters: motion transfers, posture does not.

Table 4: Task 2 linear probe on frozen features (validation macro-F1; mean±std over 3 SSL seeds). Last two columns: probe trained on the other room's SIT/STAND/WALK only.

Representation	Room A	Room B	Joint (8)	B→A sh.	A→B sh.
Handcrafted (H)	0.752	0.840	0.707	0.213	0.162
Handcrafted, denoised (H_d)	0.773	0.872	0.726	0.295	0.213
Random-weight CNN (R)	0.553 ± 0.021	0.887 ± 0.025	0.618 ± 0.009	0.198 ± 0.052	0.239 ± 0.023
SSL, raw log-amplitude (S_{raw})	0.872 ± 0.029	0.987 ± 0.004	0.931 ± 0.006	0.242 ± 0.063	0.100 ± 0.039
STRIDE-SSL (S)	0.669 ± 0.014	0.931 ± 0.010	0.740 ± 0.016	0.159 ± 0.056	0.232 ± 0.098

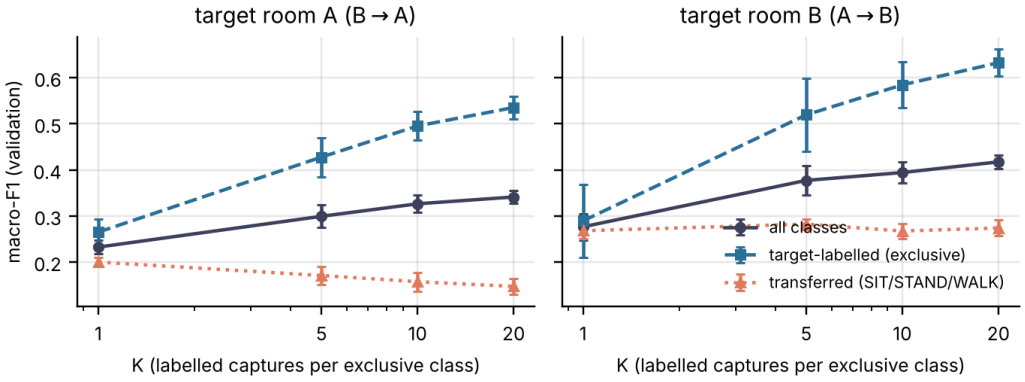


Figure 5: Task 3 label efficiency (validation; mean and 95 % CI over seeds×episodes): all classes, target-labelled (exclusive) classes and transferred shared classes.

Task 3. Table 5 and Fig. 5 give the headline. Target-labelled classes improve steadily with K; transferred shared classes stay well below them. Routing matters (Fig. 6): with the router the mean macro-F1 is 0.300 (target A) and 0.366 (target B) versus 0.228 and 0.275 without routing; predicted and oracle routing coincide because the router makes no validation errors. The ablation

(Table 6) shows (i) aligning room statistics is essential — without it no target window is ever assigned a shared class; (ii) the discriminative head beats prototypes; (iii) SSL features help on Room B and their combination with handcrafted features is competitive, but denoised handcrafted features are as strong overall; (iv) SAPT's shared-anchored statistics and balanced OT improve shared-class F1 for target A at the cost of exclusive classes, so they do not win on the combined metric — our label-shift hypothesis is only partly supported.

Denoising is the largest single gain. With the headline method and identical handcrafted features, computing them on the denoised $\hat{D}$ instead of D raises shared-class F1 from 0.051 to 0.169 (target A) and from 0.035 to 0.273 (target B), and overall macro-F1 from 0.239 to 0.300 and from 0.225 to 0.366.

Why shared-class F1 falls as K grows. More support captures make the exclusive classes broader in feature space; ambiguous target windows of the shared classes, which are only described by the *other* room's data, are increasingly absorbed by the better-described exclusive classes (Fig. 5). Overall macro-F1 still rises with K because the exclusive classes improve faster. Per class (Table 7), transfer is best for SIT in target B and worst for STAND and WALK, which are swapped across rooms (Fig. 7).

What SSL buys. Within a room, SSL is the best representation by a wide margin (Table 4). Across rooms, its advantage depends on the input: S_{raw} (static profile kept) transfers poorly, because it relies on room-specific multipath, while S (denoised dynamics) is the best single representation for target B. Neither beats denoised handcrafted features on target A, where the source room (B) offers only three shared classes recorded in noisy sessions.

Table 5: Task 3 headline (room-align+LR, H_d features), validation, mean±std over runs.

Dir.	Metric	K=1	K=5	K=10	K=20
B→A	macro-F1 (all)	0.233 ± 0.022	0.299 ± 0.038	0.327 ± 0.029	0.341 ± 0.021
B→A	macro-F1 exclusive	0.265 ± 0.044	0.428 ± 0.065	0.495 ± 0.047	0.535 ± 0.038
B→A	macro-F1 shared	0.200 ± 0.016	0.171 ± 0.029	0.158 ± 0.031	0.147 ± 0.025
B→A	balanced acc.	0.228 ± 0.020	0.319 ± 0.037	0.360 ± 0.030	0.387 ± 0.026
A→B	macro-F1 (all)	0.277 ± 0.044	0.377 ± 0.049	0.394 ± 0.035	0.418 ± 0.023
A→B	macro-F1 exclusive	0.290 ± 0.122	0.520 ± 0.122	0.585 ± 0.076	0.633 ± 0.045
A→B	macro-F1 shared	0.268 ± 0.027	0.283 ± 0.018	0.268 ± 0.025	0.274 ± 0.026
A→B	balanced acc.	0.284 ± 0.036	0.374 ± 0.049	0.401 ± 0.037	0.435 ± 0.024

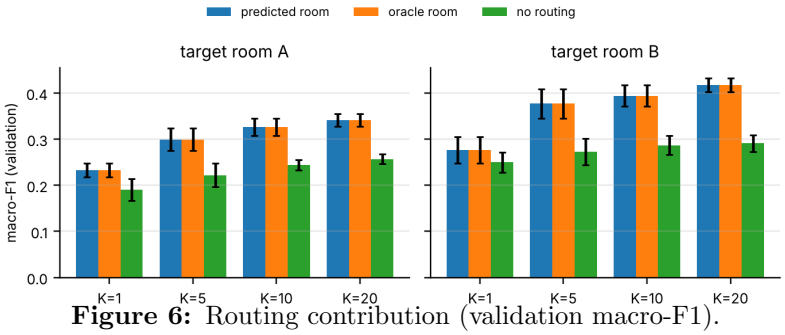


Figure 6: Routing contribution (validation macro-F1).

Table 6: Task 3 ablation: validation macro-F1 averaged over $K \in \{1, 5, 10, 20\}$ and all runs, shown as target A / target B. Rows: adaptation variant; columns: frozen features (H = handcrafted on D, H_d = on denoised $\hat{D}$, S = STRIDE-SSL, $\mathrm{S} + \mathrm{H}_d$ = concatenation).

Variant	H	H_d	S	$\mathrm{S} + \mathrm{H}_d$
LR-naive	0.220 / 0.210	0.254 / 0.241	0.199 / 0.219	0.242 / 0.220
strict(no-unlabelled)	0.172 / 0.190	0.270 / 0.282	0.203 / 0.208	0.232 / 0.214
no-routing	0.137 / 0.181	0.254 / 0.267	0.177 / 0.198	0.184 / 0.196
room-align	0.176 / 0.136	0.301 / 0.313	0.183 / 0.239	0.191 / 0.226
room-align + LR	0.239 / 0.225	0.300 / 0.366	0.224 / 0.409	0.250 / 0.411
SAPT-UDA	0.171 / 0.140	0.279 / 0.276	0.220 / 0.244	0.216 / 0.226
SAPT-UDA + LR	0.215 / 0.228	0.264 / 0.302	0.224 / 0.330	0.236 / 0.333
SAPT-OT	0.188 / 0.142	0.298 / 0.312	0.251 / 0.230	0.281 / 0.230
SAPT-OT + LR	0.188 / 0.155	0.301 / 0.326	0.291 / 0.233	0.310 / 0.237
room-OT + LR	0.232 / 0.182	0.301 / 0.323	0.215 / 0.246	0.231 / 0.246
shared-align-norefine	0.178 / 0.196	0.286 / 0.277	0.186 / 0.279	0.202 / 0.289
SAPT-UDA + LR-ST	0.209 / 0.195	0.253 / 0.285	0.218 / 0.305	0.225 / 0.300
SAPT-strict + LR	0.224 / 0.199	0.261 / 0.248	0.175 / 0.231	0.211 / 0.230
no-routing + LR	0.198 / 0.196	0.228 / 0.275	0.186 / 0.202	0.211 / 0.210

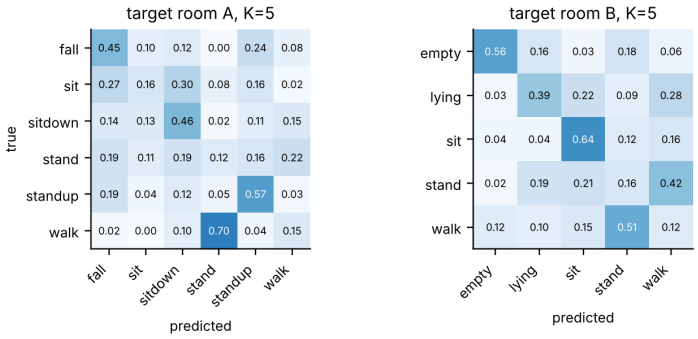


Figure 7: Confusion matrices at K=5 (accumulated over runs, row-normalised); per-class P/R/F1 in Table 7.

Table 7: Per-class precision / recall / F1 at K=5 (headline configuration, accumulated over runs).

Target	Class	P	R	F1
A	fall (exclusive)	0.36	0.45	0.40
A	sit (shared)	0.30	0.16	0.21
A	sitdown (exclusive)	0.35	0.46	0.40
A	stand (shared)	0.13	0.12	0.13
A	standup (exclusive)	0.44	0.57	0.50
A	walk (shared)	0.23	0.15	0.18
B	empty (exclusive)	0.73	0.56	0.63
B	lying (exclusive)	0.44	0.39	0.41
B	sit (shared)	0.51	0.64	0.57
B	stand (shared)	0.15	0.16	0.15
B	walk (shared)	0.12	0.12	0.12

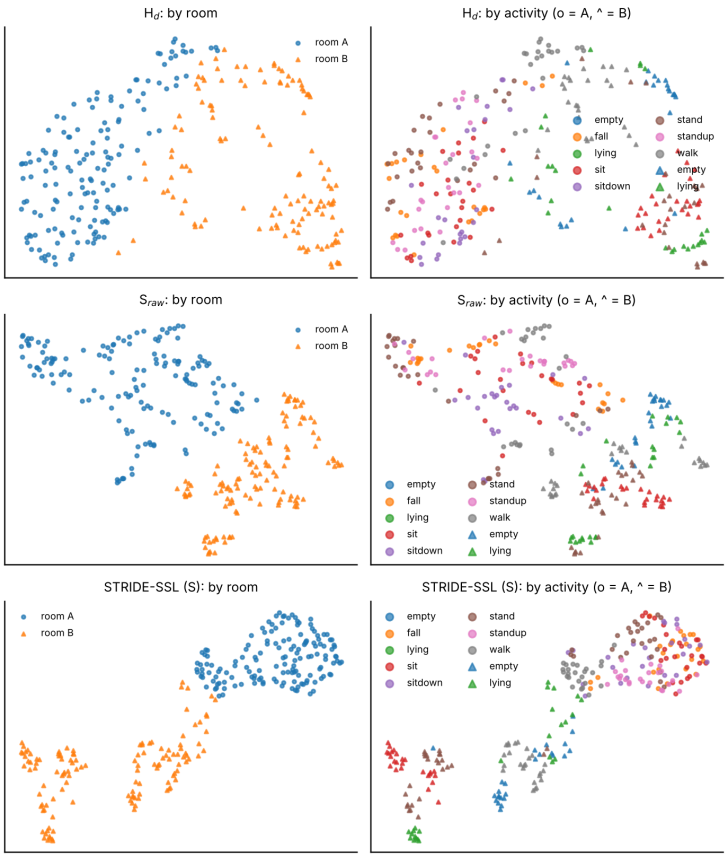


Figure 8: UMAP of validation embeddings, coloured by room (left) and activity (right). Qualitative only.

6 Reproducibility and leakage controls

The repository regenerates every artefact from the raw data with numbered scripts (`split_exp3`, `10–11` pipeline, `20/30/31/40` tasks, `50/60` figures and report numbers); every table in this report is generated by a script. Controls enforced in code and tests (18 tests): split disjointness and 25 validation captures per class; Room-B validation drawn only from the documented filename patterns; windows are exactly $(150, 256)$ float32 with no repeated padding rows; window ids are a random permutation (room not recoverable from the id); the deployment artefact contains development captures only; `run_episode.py` is smoke-tested on raw PCAP and CSV files and writes `window_id, room, label, confidence`. Only capture identity is used in SSL; validation windows are never passed to any `fit`. The SSL input scale, the harmonisation level, the handcrafted PCA and all room statistics are fitted on development data.

7 Discussion and limitations

- **Room vs session.** Rooms differ in sampling rate and noise floor as well as multipath; the router (and any embedding) exploits all of them. With two rooms we cannot separate environment from session.
- **What transfers.** Motion level (WALK vs static) transfers once noise is removed; SIT vs STAND does not, because posture is encoded mainly in the room-specific static profile. The confusion matrices also show WALK↔STAND swaps across rooms, which we attribute to session noise and to the Experiment-2 activity zones (WALK was recorded in its own zone [1]).
- **Our own pipeline choices** (rejecting 20 MHz frames, full-capture windows, interpolation of the ∼60 Room-A observations) are justified by measurements but are choices. The Room-B sampling rate cannot be recovered without timestamps; our factor of about four rests on WALK motion statistics and is only used descriptively. A full Room-B window therefore spans more

time than a Room-A window.

- **Protocol.** Results are capture-disjoint, not subject-disjoint, on two rooms only; validation guided the choice among reported variants, so headline numbers are optimistic validation results, not test accuracy. The original paper's random 80/20 results are not comparable.

Contribution statement

Solo group: all work — data engineering, modelling, experiments and report — was carried out by the single group member, with AI assistance as disclosed below.

AI usage

[Author to review and adjust to reflect actual use.] Claude (Anthropic) was used extensively as a coding and analysis assistant: implementing the pipeline and experiment scripts, running exploratory analyses of the raw data, and drafting this report. All reported numbers are produced by the committed scripts and can be regenerated from the raw data; no result was typed by hand.

Software and data

Dataset: CSI Human Activity, IEEE DataPort [1]. Libraries: NumPy, SciPy, pandas, scikit-learn, PyTorch, matplotlib, umap-learn [8]. Code and reproduction steps: repository README.

References

[1] J. Schäfer, B. R. Barrsiwal, M. Kokhkharova, H. Adil, J. Liebehenschel. Human activity recognition using CSI information with Nexmon. *Applied Sciences* 11(19):8860, 2021; dataset on IEEE DataPort, "CSI Human Activity".

[2] F. Gringoli, M. Schulz, J. Link, M. Hollick. Free your CSI: a channel state information extraction platform for modern Wi-Fi chipsets. *WiNTECH*, 2019; https://github.com/seemoo-lab/nexmon_csi.

[3] M. Gavish, D. L. Donoho. The optimal hard threshold for singular values is $4/\sqrt{3}$. *IEEE Trans. Inf. Theory* 60(8), 2014.

[4] T. Chen, S. Kornblith, M. Norouzi, G. Hinton. A simple framework for contrastive learning of visual representations. *ICML*, 2020.

[5] J. Snell, K. Swersky, R. Zemel. Prototypical networks for few-shot learning. *NeurIPS*, 2017.

[6] M. Cuturi. Sinkhorn distances: lightspeed computation of optimal transport. *NeurIPS*, 2013.

[7] Y. Hu, V. Gripon, S. Pateux. Leveraging the feature distribution in transfer-based few-shot learning. *ICANN*, 2021.

[8] L. McInnes, J. Healy, J. Melville. UMAP: uniform manifold approximation and projection. arXiv:1802.03426, 2018.